Budgeted Verification for Secondhand Valuation: Information Limits, Recovery-Path Effects and Audited Evidence

Abstract

Strategically missing fields complicate secondhand valuation when verification is budgeted and naturally absent information may remain unavailable after a charged query. We study the limits and effects of verification under this public-information boundary. In a single-recoverable-field family, we derive a tight trade-off between uniform honest precision, query capacity and worst expected concealment gain. An implementable marginal oracle supplies a finite reference, while a context-aware comparator prevents restricted policy libraries from creating an artificial solver advantage. A recovery-path decomposition shows how partial recovery can raise a non-monotone predictor's quote. A verification-history cap preserves truthful quotes and weakly reduces unilateral gain for fixed peer reports, but it can worsen attacked error or query failures after responses are recomputed. We then separate initial objective, acquisition, feedback and allocation in a ten-seed experiment on one synthetic and two advertised vehicle-price cohorts. Independent audits cover 4,320 experimental cells, 4.32 million repeated outcome records and reconstructed statistical contrasts. At the prespecified 10% capacity, all 36 feedback comparisons increase mean inflation; 34 unadjusted intervals exclude zero adversely. A robust initial objective improves all twelve MUCars inflation comparisons, whereas no joint feedback–allocation intervention passes the declared improvement gate. These results support information-consistent, multiobjective auditing rather than a universal valuation-method advantage. The intervals condition on fixed training records, and no equilibrium or training-population guarantee is asserted.

Keywords: strategic missingness; secondhand valuation; budgeted verification; incentive auditing; factorial evaluation

1 Introduction

Secondhand valuation must use incomplete seller-provided descriptions while allowing for deliberate withholding. A vehicle with no recorded mileage and a vehicle whose mileage was suppressed can present the same public report. A platform may request additional evidence, but available verification capacity is finite and a naturally absent field may remain unavailable after a paid attempt. This study models the resulting interaction between an itemwise price predictor and a batch-coupled verification policy. The outcome is an advertised price prediction, rather than a transaction price or a transfer mechanism.

A query need not reduce the incentive created by an incomplete report. Consider an illustrative nonmonotone predictor whose quotes are 10 for a truthful record, 12 after two fields are withheld, and 14 after only one is recovered. Successful partial recovery then raises the quote. These illustrative quote units are not empirical measurements. They expose a distinction between obtaining information and using the recovered record to reduce a seller's expected gain. Moreover, changing one report can displace another listing from a hard-capacity query set. Information value evaluated in isolation does not capture that competition.

Strategic withholding has established regression and classification treatments [9, 5]. Audits and partial-verification mechanism design likewise have established incentive guarantees under their respective action and outcome models [4, 6]. These results motivate verification, but they do not imply that an error-reduction acquisition score, a fixed library of query rules, or a finite-round feedback-training implementation will improve continuous valuation after failed-but-charged recovery. Ordinary active feature acquisition also targets predictive information rather than this strategic quotation objective [12].

We study two complementary questions. First, how does a uniform honest-precision requirement limit the strategic gain that a hard verification budget can remove when natural and strategic absence are observationally identical? Second, which implemented training and allocation choices actually account for changes in valuation outcomes?

A finite oracle enforces common decisions for identical public reports. A recovery-path decomposition separates the initial report, local recovery and capacity competition. A history-dependent cap provides a restricted unilateral-gain guarantee. A matched factorial experiment separates initial objective, acquisition score, policy feedback and allocation.

The contribution is a set of bounded analytical and empirical findings. Section 4 characterizes a tight precision–incentive frontier in a single-recoverable-field family and supplies an implementable marginal reference. Section 5 identifies a recovery-path failure and states the cap's fixed-peer guarantee. Sections 6 and 7.1 report a fully audited, ten-seed, three-source factorial study. At its prespecified 10% capacity, all 36 feedback comparisons increase mean inflation; a robust initial objective instead helps consistently on MUCars. Positive interaction effects can represent less harm rather than improvement. These contributions concern the stated information boundary and implementation; they do not claim a new general linear-programming method, universal method superiority or a market-equilibrium guarantee.

2 Related work

2.1 Strategic withholding and predictive missingness

Zhang et al. studied strategic missingness in regression through support-margin imputation [9]. Krishnaswamy et al. characterized truthful classifiers and constructed predictors resistant to feature withholding [5]. These works already establish that natural absence and deliberate concealment can have the same report. We therefore do not claim that continuous prediction or ambiguous missingness defines a new problem. Our analytical question adds a uniform honest-quote error constraint and a hard query budget. Removing the incentive by changing predictions may violate that precision constraint, which is distinct from average mean absolute error.

2.2 Verification, randomization and information acquisition

Estornell et al. minimize incentives through budgeted audits, including threshold and top-k settings [4]. Zhang et al. study automated classification mechanisms with partial verification, including randomized decisions and combinatorial costs [6]. Marginal-probability representations are established in security-game optimization [3]. The finite oracle here specializes those tools to public report classes and continuous quote recovery, with failed queries charged against capacity. Its complete finite coverage is a certificate within the declared world universe, not a guarantee for unseen types. Active feature acquisition addresses budgeted information selection [12]; its predictive objective differs from the signed unilateral quotation difference audited here.

2.3 Strategic feedback and contextual policy classes

Performative prediction analyzes retraining when deployment changes the data [10]; strategic empirical risk minimization differentiates through modeled user responses [11]. Contextual games use public side information to condition decisions [7]. Joint score and audit design is also explicit in recent audit-risk work [8]. We do not transfer their convergence conditions to our nonconvex, finite-round training loop or introduce randomization as a new technique. Instead, we test a frozen-policy implementation under matched fitting budgets. Our constructed policy-library separation also includes a stronger small context-aware comparator that closes the gap. This prevents a restricted library from being mistaken for evidence that dynamic weights or an LP solver are intrinsically necessary.

3 Problem formulation

3.1 Reports, queries and seller utility

A batch contains $n = 100$ listings. Listing i has a latent recorded feature vector x_i, including four optional action fields: mileage, transmission, fuel type and body condition. Let M_i be its naturally absent fields. A seller chooses a concealment set s_i containing at most two available action fields. Its public report $r_i(s_i)$ replaces fields in $M_i \cup s_i$ with missing values. Available values cannot be fabricated, changed or recovered from public identifiers. A missingness indicator does not disclose membership in M_i.

The platform observes the whole report batch $r = (r_1, \ldots, r_n)$, locks the reports and chooses a possibly randomized allocation. It may query at most one publicly missing field per selected listing, with at most $B = \lfloor bn \rfloor$ queries on every realized draw. A query of a concealed recorded field recovers that value; a query of a naturally absent field fails. Both spend one unit. A fixed itemwise model produces log quotes before and after the selected query. Scores and allocations at deployment may use public reports and model outputs, but not the latent recoverability indicator or the advertised target.

Seller utility is the expected log quote minus concealment cost. Main empirical costs are zero; all randomization occurs after reports are locked. Evaluation uses finite response search and explicitly checks every legal single-seller report against fixed peer reports, reallocating the batch each time. A returned profile with small remaining regret is a local deviation diagnostic, not a certificate of a globally selected market equilibrium. Predictions do not set a transaction price, and buyer preferences and bilateral trade are outside the model.

3.2 Predictors, public scores and comparator identity

Suppression-aware valuation (SAV) combines an augmented price predictor, a suppression acquisition head and simulated query-feedback training. The original static-robust comparator changes the prediction training objective without that feedback procedure. Counterfactual deviation-constrained mixture verification (CDM) instead fixes the price model and calibrates probabilities over complete budget-feasible query rules. Its calibration objective constrains stored own-report/peer-profile rows, not all unseen types. The original library contains twelve rules and five calibration peer-profile families; expanding the finite library does not itself establish out-of-sample incentive compatibility. These original candidates provide the context for the matched experiment, whose newly trained initial-objective factor must not be renamed SAV or static robust.

For training targets, let q_s be a cross-fitted quote under report state s, $q_{s\setminus f}$ the quote after

recovering field f, q^H the honest quote, and y the log advertised price. The three target quantities are

$$v_{sf}^{\text{sup}} = [q_s - q^H]_+ - [q_{s\setminus f} - q^H]_+, \quad v_{sf}^{\text{ben}} = (q_s - y)^2 - (q_{s\setminus f} - y)^2, \quad v_{sf}^{\text{risk}} = (q_s - y)^2.$$

A common multioutput ExtraTrees head learns these quantities from encoded public inputs [14]. Honest records and targets construct training labels only; deployment does not directly substitute them into the scores. Each policy chooses its highest-scoring publicly eligible field and queries only positive scores. Top-B ranks these per-listing scores, using the declared stable tie rule. The score-minimax control instead balances positive residual scores through query marginals. It solves a score surrogate, not the true unilateral incentive problem.

3.3 Evaluation quantities

For one returned report, let q^H be its model's honest log quote, y the log-transformed advertised price, q_0 the no-query quote, q_1 the quote conditional on the selected query, and p its marginal query probability. Here $y = \log(1 + \text{price})$ and $\mu = (1-p)q_0 + pq_1$. The principal inflation quantity is $[\mu - q^H]_+$; it compares the model with itself, rather than with the advertised target. The separate realized-positive quantity is $(1-p)[q_0 - q^H]_+ + p[q_1 - q^H]_+$. Honest price error is $|\exp(q^H) - \exp(y)|$, and attacked price error is

$$(1-p)|\exp(q_0) - \exp(y)| + p|\exp(q_1) - \exp(y)|.$$

The subtraction of one cancels inside each absolute difference. Taking an absolute error after averaging the quotes would define a different metric. Failed-query probability is p times the indicator that the selected field cannot be recovered; such an attempt still consumes capacity. Signed log bias is $\mu - y$, while expected squared log error averages the two squared endpoint errors before aggregation.

For the unilateral audit, hold peers' returned reports fixed and enumerate all feasible reports of one seller, reallocating the whole batch for each. Writing $u(r)$ for that seller's expected log quote minus concealment cost, the deviation envelope is $[\max_r u(r) - u(r_{\text{honest}})]_+$; remaining regret instead subtracts the utility of the returned report. Thus a small regret does not establish a small gain over honesty. Main empirical concealment costs are zero. All quantities are aggregated over the retained listings; nonconverged responses remain in the analysis. The finite oracle's constructed quote units must not be interpreted as empirical log-price effects.

4 Information limits and finite reference mechanisms

4.1 Public-information precision frontier

Takeaway. A finite complete oracle can distinguish a policy-library restriction from the best value achievable under the same public information.

Let o be the complete observable batch signature and $A_B(o)$ its feasible complete nonadaptive allocations, each spending at most B queries. Allocate probability z_{oa} to action a, sharing the same variables across latent worlds with the same signature. For each legal deviation d, let $G_d(z)$ be the expected deviated quote minus the truthful quote, with the entire batch reallocated on both sides. The finite oracle solves

$$\min_{z,t} t \quad \text{s.t.} \quad G_d(z) \le t \ (\forall d), \qquad \sum_{a \in A_B(o)} z_{oa} = 1, \quad z_{oa} \ge 0, \quad t \ge 0.$$

This is exact only for the declared complete finite world/report/action universe. Its numerical primal/dual certificate uses tolerance 10^{-8}. It is not a guarantee over unseen types or general adaptive query trees.

For the frontier, we optimize both verification and the itemwise quote function subject to the precision constraint; a fixed empirical predictor need not attain the bound. Consider a separate alias family with an all-natural-missing world and worlds in which only listing i has a recoverable low-value field, lowering its target by $\Delta_i > 0$. Withholding it produces the same public batch. Require verification-neutral quotes (same final record, same quote), and uniform absolute error at most η on all specified honest reports. Set $a_i = (\Delta_i - 2\eta)_+$. The attainable worst positive expected gain is

$$T^*(\eta, B) = \min_{0 \le p_i \le 1,\ \sum_i p_i \le B} \max_i a_i(1 - p_i). \tag{1}$$

For the nonzero interior case, $\sum_{i:a_i>0}(1 - T^*/a_i)_+ = B$.

Proof and boundaries. Couple queries by their private random seed on the all-failure transcript. Let p_i be the probability of querying i. Hard capacity gives $\sum_i p_i \le B$. By verification neutrality and the fixed itemwise quotation rule, unsuccessful queries do not change the quote. Until i is queried, its hidden low-value world has the same transcript; without recovery, the honest-precision constraint leaves a quote gap at least $\Delta_i - 2\eta$. This proves the lower bound, including adaptive policies in this single-recoverable family. Choose quote shifts $e_i = \min(\eta, \Delta_i/2)$ toward one another and marginals $p_i = (1 - T^*/a_i)_+$. Systematic interval sampling realizes these marginals with at most B queries per draw and attains the bound. This does not follow from an average MAE bound. It does not imply zero adaptivity benefit for failed-query counts or multi-field worlds.

What the frontier excludes. Let $k_\eta = |\{i : \Delta_i > 2\eta\}|$. Equation (1) implies $T^*(\eta, B) = 0$ if and only if $B \ge k_\eta$: each strictly positive residual gap needs unit query probability. In particular, at $B = 0$ zero gain requires $\eta \ge \frac{1}{2}\max_i \Delta_i$. For n identical gaps Δ, the frontier simplifies to

$$T^*(\eta, B) = (\Delta - 2\eta)_+ \left(1 - \min\{B/n, 1\}\right).$$

The lower bound follows because some marginal is at most B/n; equal marginals attain it. Thus honest precision, query capacity and strategic gain cannot be specified independently in this alias family.

This is compatible with existing truthful predictors. Natural-versus-strategic missingness ambiguity is explicit in [5], whose Theorems 1 and 2 give minimum-cut optimality and a MAX-ensemble characterization for binary classification. A predictor can eliminate withholding gains by changing its predictions, while violating the uniform error requirement imposed here. Our lower bound restricts that escape route; it does not contradict truthful classification or establish impossibility under average MAE alone. History-dependent quote rules satisfy this frontier only when they also meet its verification-neutrality and uniform-precision assumptions on the specified family. An arbitrary history cap cannot be placed inside that mechanism class without checking these conditions.

A two-mode constructed example at $B = 1, \eta = 0$ yields 72/17 for the public oracle, 72/13 for the best global mixture of the specified eight rules, and 8 for risk ordering. These are artificial linear quote units, not empirical log-price effects. All 32 budget/precision cases passed a separate auditor.

4.2 Compact implementability and contextual comparators

Takeaway. Item-local utility permits an exact compact oracle; enlarging a globally mixed priority library need not remove its contextual limitation.

Let y_{oif} be the probability of querying field f of item i in public report class o. Impose

$$y_{oif} \geq 0, \qquad \sum_{f} y_{oif} \leq 1, \qquad \sum_{i,f} y_{oif} \leq B,$$

and zero probability on publicly ineligible fields. Every declared unilateral deviation has the affine form $G_d(y) = b_d + \sum_{o,i,f} c_{doif} y_{oif}$ when its realized utility depends only on its own queried field or no query. Coefficients include both truthful and deviated report classes. Compatible latent worlds must share the same y variables.

Proposition 1 (Compact finite oracle)**.** *For item-local utilities, nonadaptive queries, unit query costs and at most one queried field per item, minimizing* $\max\{0, \max_d G_d(y)\}$ *over the above constraints has the same value as the complete-allocation lottery oracle. Every feasible marginal solution is implementable under a hard budget on every draw.*

Proof. Every complete-allocation lottery induces feasible y. Conversely, fix a public class and set $p_i = \sum_f y_{if}$. Arrange consecutive intervals of lengths p_i. A private uniform $U \in [0, 1)$ selects half-open intervals containing some $U + k$, for integer k. Each interval has length at most one and the total length is at most B, so at most B items are selected; item i has selection probability p_i. For $p_i > 0$, conditional on selection, draw its field with probability y_{if}/p_i. Thus every field has exactly the prescribed marginal and all affine expected deviation payoffs are preserved. □

With O public classes, n items and F fields, this representation uses at most $OnF + 1$ variables rather than one variable per complete allocation. It does not remove counterfactual-world or report enumeration, preserve arbitrary joint event probabilities, or cover adaptive queries and nonunit costs. The history cap is item-local and therefore admits the same reduction. Marginal formulations and their implementability limitations are established in security games [3]; the proposition gives the specific implementation argument needed for this valuation audit, not a claim to invent marginal optimization.

Independent checks match compact and complete-enumeration values on all 32 original market fixtures and 24 multi-field affine fixtures. One larger algebraic fixture has 100 items, 4 fields, 8 public classes, 96 payoff rows and budget 10. Its compact LP has 3,201 variables, versus 18,662,296,388,649,543,921 complete allocations per class. All 57 numerical primal/dual certificates pass. The repaired sampler has 3,200 exactly checked rational field marginals across 111 interval pieces; 16,000 numerical draws respect hard capacity. The large full-enumeration solver was not run, so this is a representation comparison, not a measured speedup.

The contextual-comparator result is a limitation of rule information, not an LP-versus-heuristic theorem. In the constructed family, a mixture over all context-insensitive priority rules has an unbounded capacity-related gap against context-conditioned allocation. A fixed mixture of only $B + 1$ context-aware rules closes that gap exactly. Appendix A gives the construction, proof and exact witnesses. Both comparators must be retained when interpreting the finite oracle.

5 Recovery paths and a conditional history cap

5.1 Signed mechanism decomposition

Takeaway. Freezing truthful queries cannot separate recovery from capacity in this replay model; use a capacity-relaxed public query instead.

For each fixed peer profile and evaluated mixture, retain its saved maximizing own report s^*. Let q^H be the truthful quote, q_h the unqueried reported quote, q_L the quote after the same rule's highest-scoring eligible query with capacity relaxed, and q_A the hard-budget quote. Ineligible or nonpositive scores still cause no query in the relaxed reference. Then

$$E = q_h - q^H, \qquad R = \mathbb{E}[q_L] - q_h,$$
$$C = \mathbb{E}[q_A] - \mathbb{E}[q_L], \qquad G = E + R + C.$$

All components are signed and evaluated at the same s^*. Mixture expectation precedes report maximization. The relaxed query is a diagnostic, not a feasible deployment policy. The original truthful-query anchor would give zero recovery: that query targets a naturally absent field, which remains absent when another field is withheld. Its remaining term would conflate recovery and capacity.

Across all nine source/seed pairs and both price models at 10% capacity, 360,000 component rows were independently reconstructed by complete stable sorting of each hypothetical batch. The audit checked 11,343,240 legal rule–world allocations and rejected four output mutations. Eighty-one saved winners differed from freshly computed argmax labels only through floating ties; the largest payoff gap was 6.94×10^{-18}. Original winners were retained. The initial argmax-identity failure is preserved in a separate run.

In the illustrative 10/12/14 example, $E = 2$ and $R = 2$. If the item receives its partial query with probability p, then $C = -2(1-p)$ and $G = 2+2p$. The cap leaves the reported quote at 12 on both branches, reducing this gain to 2 rather than making honesty optimal. Positive R can occur: hiding two fields and recovering one can move a nonmonotone estimator to an intermediate record with a higher quote. This motivates the following developmental mechanism experiment. It is not evidence that all original SAV advantages or failures have been causally explained.

5.2 History cap and fixed-peer guarantee

Takeaway. Removing quote increases along the realized recovery path gives a conditional unilateral-gain guarantee without changing truthful quotes.

For every realized allocation, use

$$q_{\mathrm{cap}}(r, a) = \min\{q(r, \varnothing), q(r, a)\}.$$

Apply this operation before averaging over a policy mixture. The underlying public scores, query budget, model and mixture weights are fixed.

Proposition 2. *If truthful queries cannot alter the fixed itemwise quote, then for any fixed peer profile and any fixed policy mixture the cap preserves the truthful quote and weakly reduces the worst expected unilateral gain over legal own reports.*

Proof. Pointwise $q_{\mathrm{cap}}(s, a) \le q(s, a)$, whereas $q_{\mathrm{cap}}(0, a) = q(0, a) = q^H$ for every realized action. Subtracting the same truthful quote, taking expectations and then maximizing over legal s preserves the inequality. The null deviation yields zero. □

The mechanism deliberately uses verification history. It lies outside the verification-neutral class in Equation (1), so it cannot be used to claim a violation of that bound. Equal final records can have different prices under the cap. The guarantee concerns fixed peers and scores; a new seller-response profile can change allocations and outcomes. Lower quotes need not have lower absolute pricing error.

The full comparison includes both estimators and all six capacities: 108 cases, 5,671,620 legal report rows and 2,160,000 entry–method comparisons. A separate auditor reconstructs absolute quotes rather than reusing the signed-gain cap calculation. All truthful quotes are unchanged and all fixed-profile envelopes weakly decrease within tolerance 10^{-10}. Risk has zero improvement in the primary fixed-profile table in Appendix B and is retained as a strong reference. These are additional conditional diagnostics, not new ten-seed training experiments.

6 Experimental design and independent audits

6.1 Cohorts and source boundaries

Takeaway. The evidence concerns advertised-price prediction on fixed retrospective cohorts and explicitly modeled seller responses.

JUCars-2024 version 2 contains Jordanian listings from 2024 and was deposited in 2026 [1]; MUCars-2024 version 2 contains Moroccan listings from 2024 [2]. Both sources are distributed under CC BY 4.0. Recorded prices are advertised prices, not completed transaction prices. We evaluate native currencies separately. The synthetic market follows the saved generator rather than an empirically identified transaction model. Before the present caps, the processed training/evaluation splits contain 4,774/1,374 JUCars records and 39,244/10,672 MUCars records. Identifier hashing caps training at 5,000 and evaluation at 1,000 records. Synthetic generation uses fixed seeds 73,101 and 73,102 for 6,000/1,500 records before the same caps. Fitting seeds vary training randomness, not the source cohorts or generator. These validation sources were inspected during earlier development and are retrospective, rather than fresh confirmatory tests.

Original advertisement identifiers are unavailable; inherited non-price fingerprints provide duplicate control without certifying independent vehicles. Targets, identifiers, source indices and declared direct reveal aliases are excluded from predictor inputs. Correlated covariates remain, so concealment need not remove all information about a field. Naturally absent fields cannot be concealed by a seller or successfully recovered by a query.

6.2 Matched training and evaluation

Takeaway. The full controlled experiment and four-stage audit are complete; controlled contrasts do not establish a general method advantage.

The new design crosses two initial training objectives (mask-augmented and robust), three acquisition scores (suppression, benefit and risk), two policy feedback settings, and two allocators (top-B and score-minimax). The initial objective factor is a new controlled intervention, not a relabeling of the original SAV and static-robust checkpoints. Each cell uses the same backbone architecture, six initial residual epochs and six subsequent epochs in two rounds, with matched record orders and head-refresh schedules. No-policy-feedback fits are shared across their six acquisition/allocation aliases, giving fourteen unique fits for twenty-four cells. The implemented sequence is:

1. For each of three outer folds, exclude its records from the complete training loop. Split the remaining records into estimator and head-guidance sets, using a guidance fraction of 0.3.

Fit the augmented backbone and each six-epoch initial residual objective.

2. Clone each initial model into one no-policy-feedback fit and six score/allocator-specific feedback fits. Run two three-epoch robust-update rounds. Refit the guidance head after each round, retaining identical record orders across the matched cells.

3. At the start of each feedback epoch, compute public acquisition scores and hold them fixed during that epoch. Within each minibatch, use the current detached quotes to search seller responses, with two sweeps and one start at 10% capacity. Reallocate the batch for every candidate report. Update only the residual model using the resulting frozen response and probabilities. The no-policy-feedback control instead searches valid reports with no queries; it still receives the same six robust-update epochs.

4. Predict every excluded fold with its corresponding trained model. Fit the deployment model through the same training sequence, then refit its 160-tree deployment heads using the assembled outer-fold quote targets and the deployment model's encoded inputs. These heads are used for evaluation; the deployment residual model is not updated after this final head fit.

5. Evaluate each fixed deployment model at capacities 0, 1, 5, 10, 20 and 30%, using four response starts and at most twenty sweeps. Compute expected quotes and errors analytically from the query marginals; retain every final profile and its independently checked unilateral regret.

This protocol does not refit a separate model for each evaluation capacity. Effects away from 10% therefore include transfer from the fixed training capacity. Evaluation uses stronger response search than training, but finite search still does not certify a global best-response equilibrium.

For the score-minimax control, an item's highest positive publicly eligible score is g_i. We minimize $\max_i g_i(1-p_i)$ subject to $0 \le p_i \le 1$ and $\sum_i p_i \le B$, then realize the marginals with a private systematic draw. This is a score surrogate and an established resource-allocation construction; it is not a guarantee about report-dependent unilateral incentives. Every candidate seller report therefore triggers a fresh batch allocation in the response calculation.

For a frozen within-step query marginal p, training uses the expected post-query squared error,

$$(1-p)(q_0-y)^2 + p(q_1-y)^2 = ((1-p)q_0 + pq_1 - y)^2 + p(1-p)(q_1-q_0)^2.$$

Taking squared error after averaging quotes would omit the variance term. The gain penalty instead uses the positive part of expected gain, matching the expected-utility objective. Gradients do not pass through the discrete report search, field selection or allocation probabilities; this remains a frozen-policy training surrogate.

Writing L_H for truthful squared error, L_V for mean squared error over valid report states and L_A for the squared anchor to the backbone's honest quote, the mask objective is $L_H + 0.25L_V + 0.25L_A$. The robust objective adds $0.5L_{\text{attack}} + 0.5L_{\text{gain}}$, where $L_{\text{gain}} = \mathbb{E}[(\mu - q^H - 0.02)_+]$ and L_{attack} averages the endpoint squared errors before expectation. The honest anchor in the gain penalty is detached. AdamW uses weight decay 10^{-4} and gradient clipping at norm 5. The residual has two width-64 ReLU hidden layers and a scalar output. The subsequent robust update combines truthful squared error, a quarter-weighted mean squared error over valid report states, a quarter-weighted anchor to the backbone's truthful quote, and half-weighted expected attacked error and gain penalty. The declared weights and twelve-epoch total are common across matched cells. The distinction between expected squared error and squared expected quote is necessary for correct stochastic replay training; the variance identity itself is not a new mathematical result.

For attribution, at fixed source, initial objective, acquisition score and capacity, let $L_{f,a}$ be an outcome under feedback setting f and allocator a. We report the interaction

$$D = (L_{\text{none,minimax}} - L_{\text{policy,minimax}}) - (L_{\text{none,top}B} - L_{\text{policy,top}B}).$$

For loss-like outcomes, positive D means feedback has a larger incremental benefit with minimax. Both constituent effects must also be shown: positive D alone does not establish that either feedback policy is beneficial. The fixed analysis contract contains 58 conditional factor contrasts per source/capacity, including six feedback–allocation interactions. The primary capacity therefore has 174 exploratory contrasts, all retained rather than screened for favorable directions. No individual unadjusted interval is treated as a confirmatory finding across this family.

The full experiment completed thirty source–seed pairs and 4,320 cells. The training audit reconstructed hash-ranked partitions, matched minibatch orders and 2,068,360 held-out prediction rows. It independently rebuilt all 420 deployment heads, comparing tree structure and leaf values, and rejected altered partitions, permutations and predictions. This audit verifies fit provenance and head targets, rather than replaying every optimizer step. The outcome audit checked 4,320,000 recorded outcomes and 45,372,960 legal unilateral deviations, including reallocation after each changed report. An independent statistical checker reconstructed 9,216 metric–contrast rows, validated contrast semantics and intervals, and rejected altered interaction signs and interval endpoints. The full four-stage completion receipt was issued on 13 October 2026.

Table 1 records the declared settings.

Setting	Value
CatBoost backbone boosting rounds [13]	250
Residual hidden width / learning rate	64 / 0.001
Initial epochs / robust-update epochs	6 / 6
Outer folds / guidance fraction	3 / 0.3
Acquisition-head trees / maximum depth / minimum leaf	160 / 12 / 12
Training budget / response starts / response sweeps	10% / 1 / 2
Evaluation response starts / maximum sweeps	4 / 20
Batch size / maximum concealed available fields	100 / 2
Gain-penalty tolerance / main concealment cost	0.02 / 0
Fitting seeds / crossed bootstrap repetitions	10 / 1,000

Table 1: Declared matched-experiment settings. Fitting seeds are 13, 42, 2026, 101, 211, 307, 401, 503, 601 and 701. The backbone and ExtraTrees implementations follow established methods; they are not new algorithmic contributions. All capacities share the fixed training budget.

7 Results

7.1 Matched factorial effects

Takeaway. At the prespecified 10% capacity, none of the 36 feedback comparisons reduces mean inflation. A robust initial objective helps consistently on MUCars, but positive interactions do not imply beneficial feedback or a successful combined policy.

Inference scope. Every source uses ten fitting seeds and ten shared evaluation batches of 100 listings. The same 1,000 listings are reused across seeds and methods; 4.32 million outcome records

are therefore not independent listing observations. We report 1,000 crossed seed–batch bootstrap draws and their 2.5th and 97.5th percentiles. These unadjusted exploratory intervals condition on the fixed training records. They do not cover a new training population or correct for multiple comparisons. The primary capacity supplies 174 factor contrasts and 18 additional joint-policy contrasts, each evaluated on eight metrics. All six capacities and all contrast components remain in the evidence package. Comparisons away from 10% also measure transfer from the fixed training capacity.

Feedback does not explain a beneficial effect in this design. Define a feedback contrast as no-policy-feedback loss minus policy-feedback loss, holding the initial objective, score and allocator fixed. For positive expected log-quote inflation, all 36 point estimates at 10% were negative; 34 unadjusted intervals excluded zero in the unfavorable direction. The two intervals crossing zero corresponded to MUCars with the mask initial objective and suppression acquisition. No feedback contrast passed the point gate of at least 10% inflation reduction with at most 5% honest-MAE increase. This is a negative result for the implemented two-round frozen-policy training procedure, not an impossibility result for feedback learning.

Table 2 summarizes all primary directions.

Source	Feedback harms	Harm intervals	Robust-init gains	Joint gates
Synthetic	12/12	12/12	4/12	0/6
JUCars	12/12	12/12	12/12	0/6
MUCars	12/12	10/12	12/12	0/6

Table 2: Primary-capacity factorial diagnostics. Counts refer to positive expected log-quote inflation. Feedback harms and robust-initialization gains are point-estimate directions. Harm intervals are 95% unadjusted intervals excluding zero adversely. A joint gate compares no feedback/top-B with policy feedback/minimax under a fixed initial objective and score, requiring at least 10% inflation reduction and at most 5% honest-MAE increase. Counts are descriptive across correlated contrasts, not independent trials.

Figure 1 displays the suppression-acquisition comparisons.

Initial-objective effects depend on the source. The robust initial objective lowered mean inflation in all twelve MUCars conditional contrasts. Reductions ranged from 12.43% to 33.40%, all twelve unadjusted intervals excluded zero favorably, and all twelve passed the point gate. Honest MAE also decreased in these comparisons. On JUCars, twelve point estimates favored the robust objective but only two intervals excluded zero favorably. On synthetic data, only four point estimates favored it, and six intervals excluded zero in the adverse direction. Thus the controlled initial-objective factor has a stronger source-dependent signature than a beneficial feedback effect. These are newly trained matched models, so they do not retrospectively identify the cause of the original SAV/static-robust comparison.

Figure 2 shows the no-feedback suppression comparisons.

A positive interaction can mean less harm. For synthetic data with the mask objective and suppression acquisition, the feedback–allocation interaction was $D = 0.004917$, with an unadjusted interval $[0.002378, 0.007586]$. Feedback increased inflation under both allocators: from 0.010871 to 0.018969 under top-B, and from 0.010638 to 0.013818 under minimax. The positive interaction measures the smaller increase under minimax; it does not show that minimax feedback improved

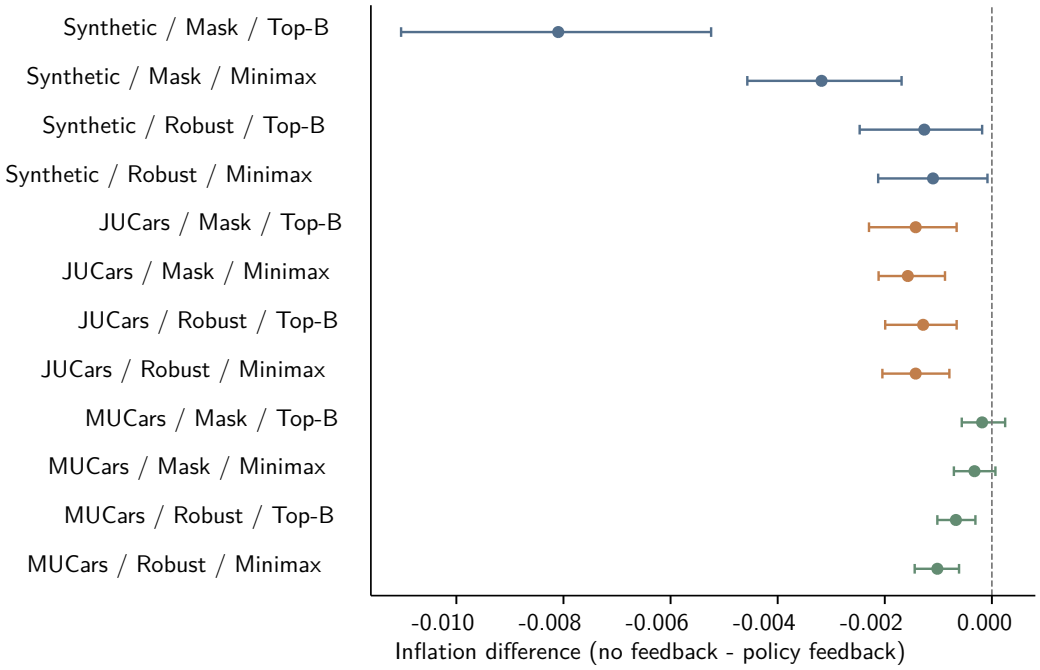


Figure 1: Policy feedback increases mean inflation in all twelve suppression-acquisition cells at the primary 10% capacity. Points are no-feedback minus policy-feedback paired differences; bars are 95% unadjusted intervals from 1,000 crossed resamples of ten fitting seeds and ten shared batches. Positive differences would favor feedback. All models use the same fixed training records; 1,000 evaluation listings are reused across seeds. The two MUCars mask-objective intervals cross zero. Other scores are retained in the full contrast data.

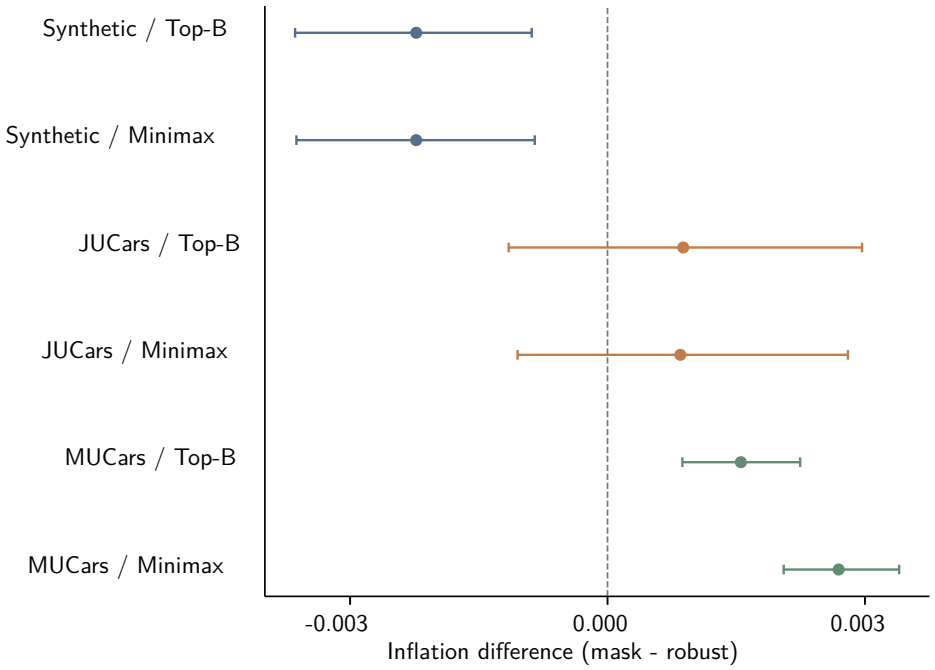


Figure 2: Initial-objective effects depend on the source. With suppression acquisition and no policy feedback, points are mask-objective minus robust-objective inflation differences at 10% capacity. Positive values favor robust initialization. Intervals and repeated-unit definitions match Fig. 1. The source labels and row positions encode cohort identity independently of color. These newly trained objectives are not aliases for the original SAV/static-robust candidates.

on its no-feedback control. Across all sources, none of eighteen joint changes from no feedback/top-B to policy feedback/minimax passed the point gate. The three favorable MUCars joint point estimates were small and their intervals crossed zero. Inflation, honest error, attacked error and failed-query effects must therefore be read jointly rather than inferred from the interaction sign.

Mechanism and practical boundary. The factorial experiment isolates implemented training and allocation choices, but does not identify why feedback increased inflation. Frozen acquisition scores, the weaker training response search and capacity transfer are possible explanations requiring separate interventions. They are not established causes. The appropriate practical comparison is the matched no-feedback control with the same initial objective, score and capacity. Choosing feedback from a positive interaction alone is unsupported. A risk score is also not a universal fallback: its feedback-trained cells are among the adverse cases. Performance on a new cohort requires renewed multiobjective validation; point Pareto flags in the present cohort are descriptive, not probabilistic dominance guarantees.

Absolute outcomes qualify an inflation-only ranking. Table 3 holds the robust initial objective and suppression acquisition fixed. Minimax feedback increased JUCars inflation from 0.010432 to 0.011855 while reducing attacked MAE from 6902.07 to 6899.03 and lowering query failures. On MUCars, the same feedback contrast increased both inflation and attacked error. Thus improving prediction error need not reduce the seller's gain over the model's own honest quote. Of the 432 source/capacity/cell settings after aggregating seeds, 37 contained a nonconverged batch; eight of these were synthetic primary-capacity settings. The maximum recorded remaining regret was 0.177513 log units. All were retained.

Source	Feedback/allocation	Inflation	Honest MAE	Attack MAE	Failure (%)
Synthetic	None/top-B	0.013101	3582.47	3655.51	2.430
Synthetic	None/minimax	0.012868	3582.47	3653.16	1.886
Synthetic	Policy/minimax	0.013967	3589.84	3652.94	2.641
JUCars	None/top-B	0.011112	6909.46	6902.00	6.330
JUCars	None/minimax	0.010432	6909.46	6902.07	5.688
JUCars	Policy/minimax	0.011855	6907.68	6899.03	5.485
MUCars	None/top-B	0.006511	27223.61	27372.61	0.380
MUCars	None/minimax	0.005371	27223.61	27363.32	0.651
MUCars	Policy/minimax	0.006390	27267.93	27441.51	0.542

Table 3: Absolute primary-capacity outcomes for the robust initial objective and suppression acquisition, averaged over ten fitting seeds and the same 1,000 evaluation listings. Inflation is positive expected log-quote inflation; MAEs use the source's own currency, without currency pooling. Failures are per listing, rather than conditional on receiving a query. These are descriptive means; the effect intervals are in the full paired-contrast data.

7.2 Original candidate comparison: a distinct experiment

The original SAV study supplies a comparator check rather than a factorial attribution. Its ten-seed extension covers 30 source–seed pairs, 1,860 cells and 1.86 million repeated outcomes. At 10% capacity, SAV inflation was 0.021175 on synthetic data, 0.014988 on JUCars and 0.007340 on MUCars; the corresponding static-robust values were 0.012084, 0.012082 and 0.006511. The static-robust-minus-SAV crossed intervals were $[-0.012355, -0.006323]$, $[-0.007025, 0.000777]$ and

$[-0.001870, 0.000210]$, respectively. These comparisons do not support consistent SAV superiority; intervals including zero also do not establish equivalence. Appendix C retains all five original comparators for each source, including strong benefit-head controls and weaker masking/no-gap variants. These original models must not be substituted for the matched factorial objectives.

Appendix E retains ordinary-benefit, equal-tuning, categorical support-margin and acquisition-conditioned neighbor controls, with their distinct fitting and adaptation assumptions.

7.3 Allocation changes on frozen models

Takeaway. Score-minimax allocation can help with a robust price model, but the same intervention can worsen inflation with a masking model. It is not a universal replacement for top-B.

This separate experiment retains the original three seeds and frozen price models. It crosses three sources, two price models, three public acquisition scores and two allocators at 10% capacity: 108 cells and 108,000 outcome rows. Seller responses are recomputed for each allocator. The independent evaluator reconstructs 1,080 method/batch cells, 1,134,324 legal unilateral deviations, model outputs, allocations and expected-error arithmetic. A second checker rebuilds all eighteen contrasts, five metrics, resampling intervals, decision gates and pointwise dominance flags from the raw records. This is an allocation intervention conditional on frozen models, not the full feedback-training study.

Table 4 retains all eighteen contrasts. Only robust-benefit and robust-suppression on MUCars meet the pre-existing point criteria of at least 10% inflation reduction and at most 5% honest-MAE increase. They reduce inflation by 13.70% and 13.05%, but increase failed-query probability by 0.060 and 0.128 percentage points. Honest MAE is unchanged in every contrast under these replay semantics. Ten of eighteen contrasts increase inflation; four have a positive lower endpoint for the inflation reduction. JUCars robust-benefit improves by 9.985%, which does not meet the 10% threshold. No threshold is relaxed after observing this near miss.

Table 4: Allocation intervention at 10% capacity. Positive inflation reduction favors score-minimax over top-B. Intervals are in log-inflation units, from 1,000 crossed resamples of three fitting seeds and ten shared batches, conditional on fixed training records; they are unadjusted. The final column is minimax minus top-B failed-query probability in percentage points, so a positive value is an adverse change.

Source	Price	Score	Reduction (%)	95% interval	Failure change
jucars	masking	benefit	0.49	$[-0.00005, 0.00027]$	0.204
jucars	masking	risk	-3.26	$[-0.00181, 0.00009]$	0.015
jucars	masking	suppression	-0.30	$[-0.00024, 0.00012]$	-0.604
jucars	robust	benefit	9.99	$[0.00065, 0.00160]$	0.815
jucars	robust	risk	0.52	$[-0.00029, 0.00040]$	-0.011
jucars	robust	suppression	8.00	$[0.00044, 0.00132]$	-0.634
mucars	masking	benefit	-0.51	$[-0.00069, 0.00013]$	-0.043
mucars	masking	risk	-36.94	$[-0.01262, -0.00726]$	-0.000
mucars	masking	suppression	-0.55	$[-0.00066, 0.00009]$	-0.041
mucars	robust	benefit	13.70	$[0.00087, 0.00128]$	0.060
mucars	robust	risk	0.13	$[-0.00024, 0.00024]$	-0.000
mucars	robust	suppression	13.05	$[0.00082, 0.00126]$	0.128
synthetic	masking	benefit	-47.57	$[-0.02905, -0.01810]$	-1.543
synthetic	masking	risk	-49.33	$[-0.02998, -0.01744]$	-1.659

Source	Price	Score	Reduction (%)	95% interval	Failure change
synthetic	masking	suppression	-22.54	$[-0.01618, -0.01067]$	-1.597
synthetic	robust	benefit	-6.65	$[-0.00193, 0.00038]$	-1.186
synthetic	robust	risk	-11.36	$[-0.00221, -0.00037]$	-0.928
synthetic	robust	suppression	2.34	$[-0.00009, 0.00085]$	-0.633

The multiobjective results qualify the apparent gains. MUCars robust-benefit and robust-suppression also reduce expected attacked MAE by 7.17 and 7.67 source-currency units, but their failed-query increases prevent Pareto dominance. JUCars robust-benefit and robust-suppression increase attacked MAE by 2.06 and 0.62 units despite reducing inflation. Only JUCars robust-risk and synthetic robust-suppression weakly dominate top-B on all four point metrics; neither has an inflation interval excluding zero. These point comparisons use a numerical tie tolerance, not a claim of practical equivalence.

Ten top-B response batches fail the convergence criterion (six synthetic masking-benefit and four synthetic masking-suppression); their maximum unilateral regret is 0.095752. The minimax responses have zero recorded final regret over the enumerated legal unilateral actions. All batches are retained. Thus the intervention includes its response-search consequences, and the observed reversals cannot be attributed solely to equilibrium allocation quality. The score objective remains a surrogate and does not prove general incentive optimality. The completed factorial study evaluates whether training feedback changes these conditional effects.

7.4 Recomputed cap responses and multiobjective trade-offs

Takeaway. The cap's inflation reductions survive the saved response heuristic in this study, but attacked error and failed-query rates can worsen. This is an empirical result, not an extension of the fixed-peer theorem.

We recomputed reports for every source/seed/estimator/capacity/rule cell, using the original four starts and up to 20 sweeps, frozen public scores and unchanged mixture weights. The uncapped baseline uses the independently audited original responses. All 432 new cells and 432,000 listing outcomes were replayed by an independent full-sort auditor. It checked 5,300 supported complete-rule batch allocations and 5,570,112 legal rule–deviation outcomes. The auditor reconstructs final quotations, currency absolute errors, query failures and remaining unilateral regret; it does not certify a globally optimal joint response or every search step.

Table 5 reports the primary robust-estimator comparison.

Across all 432 paired cells, inflation decreases in 279 and is unchanged in 153 within numerical tolerance. Attacked MAE decreases in 185 cells, is unchanged in 153, and increases in 94. Failed-query probability increases in 145 cells, decreases in 4, and is unchanged in 283. Honest MAE is unchanged throughout. These cell counts are descriptive; cells share records and fitted models.

Thirty-three capped cells contain at least one batch with remaining unilateral regret above the convergence tolerance, compared with 34 original cells. The maximum capped regret is 0.204423 log units. We retain these cases. In the primary robust benefit comparison, one synthetic seed has one nonconverged batch, with maximum regret 0.029232; the JUCars and MUCars primary benefit batches pass the specified local regret check. Therefore no result is described as a certified market equilibrium.

For the 12 primary source/rule contrasts, we compute unadjusted 95% percentile intervals from 1,000 paired resamples of the ten shared batches, first averaging the three fixed fitted-seed outcomes within each batch. These are conditional evaluation-batch intervals, not training-population intervals or 30 independent batch replicates. For benefit, the absolute inflation-reduction intervals

Table 5: Recomputed common-response comparison at 10% capacity with the robust estimator, averaged over three seeds. Positive inflation reduction is favorable; positive attacked-MAE change and failed-query change are unfavorable. Failure change is percentage points per listing, not the conditional failure/query rate.

Source	Rule	Inflation reduction (%)	MAE change (%)	Failure change (pp)
synthetic	suppression	4.82	-0.079	+0.400
synthetic	benefit	7.72	-0.147	+0.467
synthetic	risk	3.70	-0.122	+0.133
synthetic	cdm	1.07	-0.081	+0.160
jucars	suppression	9.70	+0.070	+0.333
jucars	benefit	12.44	+0.080	+0.800
jucars	risk	0.61	+0.002	+0.000
jucars	cdm	7.71	+0.018	+0.300
mucars	suppression	7.62	-0.038	+0.133
mucars	benefit	11.76	-0.030	+0.200
mucars	risk	1.11	+0.005	+0.033
mucars	cdm	4.76	-0.019	+0.067

are $[0.000388, 0.001252]$ (synthetic), $[0.001207, 0.001586]$ (JUCars), and $[0.000842, 0.000995]$ (MUCars). The JUCars and MUCars point reductions satisfy the original 10% inflation/5% honest-error criterion with a positive conditional interval, but that criterion does not constrain attacked error or query failures. In particular, JUCars attacked MAE increases by 0.080%.

Risk had no change in the earlier primary fixed-profile table, but can change under the newly computed profiles. The comparisons condition on different peer reports, so these findings are compatible. The existing fixed-profile theorem must not be used as a common-response or Pareto-dominance guarantee.

8 Discussion

Takeaway. The supported contribution is a bounded information frontier and an audited verification-path mechanism. General superiority is unsupported; the completed factorial experiment separates source-dependent initial-objective effects from adverse feedback effects.

Table 6 connects the principal claims to their evidence.

Table 6: Claim–evidence matrix. Positive comparator-minus-SAV differences favor SAV. All listed experiments are complete; bounded claims remain separate from general superiority.

Claim	Decisive evidence	Verdict and boundary
C1. A precision–incentive frontier is tight.	Restricted missingness-alias proof and 32-case finite oracle.	Supported in the stated family; uniform honest error and verification-neutral quotes required.
C2. Marginals replace complete allocation lotteries.	57 compact cases; 56 comparisons with complete enumeration.	Supported for item-local utility, unit costs and at most one field per item; marginal optimization is established prior work.

Claim	Decisive evidence	Verdict and boundary
C3. The fixed-priority library can have an unbounded gap.	26 exact cases; a fixed mixture of $B+1$ context-aware rules closes the constructed gap.	Supports a restriction on rule information; does not establish an advantage of dynamic weights or the LP solver.
C4. A verification-history cap reduces fixed-profile incentives.	Pointwise replay argument and exhaustive fixed-profile evaluation.	Supported for fixed peer reports and the stated truthful replay; not an equilibrium theorem.
C5. The cap improves every measured objective after recomputed responses.	432 cells: 94 attacked-error increases and 145 failed-query increases.	Not supported. Inflation reductions do not imply Pareto dominance.
C6. SAV consistently beats static robust at 10% capacity.	Ten-seed paired difference: synthetic -0.009091; JUCars -0.002906; MUCars -0.000829.	Not supported. Synthetic is unfavorable with an interval excluding zero; both real-source intervals include zero.
C7. Score-minimax allocation consistently improves outcomes.	108 audited cells; 18 independently checked contrasts; only two pass the point gate, while ten worsen inflation.	Not supported generally. MUCars robust gains coexist with failed-query increases; the separate factorial study does not show beneficial feedback.
C8. Matched feedback training reduces inflation.	4,320 audited cells; at 10% capacity, all 36 feedback point estimates worsen inflation and 34 adverse intervals exclude zero.	Not supported for this training procedure. This does not establish why feedback fails or rule out other feedback algorithms.

The information frontier and recovery-path findings answer different questions. The frontier limits what verification can guarantee under observationally indistinguishable missingness; the recovery analysis shows how an otherwise valid query can change the quotation adversely. Together they support auditing both the information available to an allocator and the quotation after recovery. The compact formulation supplies an exact finite reference, while the small context-aware rule library shows why a solver advantage cannot be inferred from a restricted comparator alone.

For deployment, the history cap is a conditional intervention whose attacked error and failed-query effects need to be measured alongside inflation. The source-specific SAV results similarly support local validation against static robust and simple acquisition controls. They do not justify selecting SAV as a universal default. The completed factorial experiment identifies initial-objective effects on MUCars and adverse feedback effects across all three cohorts. These controlled effects do not establish the cause of the original unmatched SAV comparison, and response-search behavior remains a possible contributor. Fitting-seed resampling alone cannot establish uncertainty over new training populations.

The study does not infer actual seller behavior from the advertised-price cohorts. Concealment sets, utility, query success and response search are stipulated. Only two real cohorts, four query fields, unit costs and batches of 100 are evaluated. New market conditions, noisy verification, multi-field queries, nonunit costs and adaptive interrogation can change the frontier or the observed ranking. These limits restrict deployment claims rather than invalidate the explicitly audited finite comparisons.

9 Conclusion

Budgeted verification has both an information limit and a quotation-path consequence. Uniform honest precision restricts the concealment gain that can be removed when natural and strategic absence generate the same report. A history cap can suppress adverse recovery paths under fixed peers, while recomputed responses expose attacked-error and failed-query costs. The matched experiment further shows that a positive feedback–allocation interaction is insufficient evidence of improvement: all 36 primary feedback comparisons increase mean inflation, even though robust initialization helps consistently on MUCars. The practical implication is to audit public information, recovery semantics and every comparison component, retaining negative results and multiobjective trade-offs. This conclusion is conditional on the implemented learning and response procedures, not a universal prescription for valuation.

Data, code and use of AI tools

The source datasets are publicly available under CC BY 4.0 through the versioned Mendeley Data records [1, 2]. All reported prices are advertisements. A private research repository contains the implementation, full six-capacity analysis tables, resampling tensors, figure source data and hash-indexed audit receipts. It is not yet a public archival release. Model weights and large per-listing outputs remain on the research machine; independent clean-machine reproduction has not been demonstrated. The source access and reconstruction instructions are included in the evidence package. External access and an archival release must be finalized before an editorial submission requiring public availability.

AI assistance was used for implementation support, experiment orchestration, analysis-code development, literature checking, figure preparation and manuscript drafting. Quantitative claims are tied to saved experimental records and independent programmatic reconstructions; AI-generated empirical values were not substituted for them. Human scientific review, author attribution and final responsibility remain to be supplied before submission. Author names and affiliations are deliberately omitted from this version.

A Contextual policy-library witnesses

A capacity-related global-mixture gap. Consider $n > B \geq 1$, a fixed core C of size B, and public contexts $j \in \{B+1, \ldots, n\}$. In context j, potential quote gaps are 2 for core items, 1 for item j, and $\epsilon = 1/(B+2)$ for every other item. These are linear toy utility units. Use the honest-exact missingness-alias worlds from the preceding construction. All items are query-eligible in the all-missing report.

The reference library contains *all* fixed item-priority orders and a risk rule using these exact gaps, but employs one context-independent mixing vector. Every priority rule selects a fixed B-subset. Risk always selects the core, because $2 > 1 > \epsilon$, without relying on a cutoff tie. Thus the global mixture has the same query marginals in every context, whereas a publicly conditioned mechanism may change them with j.

Proposition 3 (Global-mixture separation)**.** *In this family the optimal worst gains are*

$$T_{\text{global}} = \frac{2(n-B)}{2n-B}, \qquad T_{\text{context}} = \frac{2}{B+2}.$$

Their ratio is $(B+2)(n-B)/(2n-B)$. *Taking* $n = B^2 + B$ *makes the ratio* $B(B+2)/(2B+1)$, *which grows without bound with* B.

Proof. A global mixture with value t must satisfy $2(1-p_i) \le t$ for each core item and $1-p_i \le t$ for each other item, using the context in which that item is active. Summing gives $B \ge n - t(B/2 + n - B)$ and the stated lower bound. It is attained with core marginal $1 - T_{\text{global}}/2$ and other marginal $1 - T_{\text{global}}$. These sum to B and are implementable as a lottery over B-subsets, all represented by fixed priorities. In one public context, the B core items and active item imply $B \ge B + 1 - t(B/2+1)$. Set core marginal $(B+1)/(B+2)$, active-item marginal $B/(B+2)$ and all others zero to attain the bound. The low-gap unqueried items have gain $\epsilon < T_{\text{context}}$. Both constructions use the preceding hard-budget sampling argument. □

We checked rational primal/lower-bound witnesses on 26 predetermined (n, B) cases and 257,274 context–item rows. Sixteen small cases also match independently constructed complete-subset LPs for both policy classes. At $B = 20, n = 420$, the ratio is $440/41 \approx 10.73$. This is a constructed limitation of global mixing over the specified context-insensitive rule library; it does not establish that real-source CDM losses arise from this mechanism or that these bounds are first in the literature.

A stronger small-library comparator. The separation concerns the information used by the rules, not fixed mixture weights alone. Add $B+1$ deterministic context-aware rules: one queries every core item; rule i replaces core item i with the publicly active item. Give the first rule weight $2/(B+2)$ and each replacement rule weight $1/(B+2)$. The weights are fixed across contexts. Every draw uses exactly B queries, while each core item's marginal is $(B+1)/(B+2)$ and the active item's marginal is $B/(B+2)$. Thus this small library attains T_{context} without solving a new LP at deployment. The previous lower bound proves its optimality within the constructed family. It does not imply an unseen-context guarantee for learned rules on vehicle records.

An independent exact-arithmetic extension checks all 26 cases, 15,922 context–rule allocations and 257,274 context–item losses. Freezing the active item or changing the weights to uniform fails the optimum check. Consequently, this family supports a representation limitation of the original library, not an intrinsic advantage of dynamically chosen weights or an LP solver. This stronger comparator must be retained in subsequent algorithm evaluations.

B Fixed-profile cap comparisons

Table 7 retains all four rule comparisons for each source.

C Complete original-protocol ten-seed comparisons

C.1 synthetic

Takeaway. Increasing the fitting-seed count preserves the negative synthetic result against the original static robust comparator. This section uses ten seeds; preceding cap experiments retain three.

We combine original seeds 13, 42 and 2026 with predetermined seeds 101, 211, 307, 401, 503, 601 and 701 under the same original protocol. All ten synthetic checkpoints pass the source-binding, split, model-receipt and recorded-outcome checks: 620 method/capacity cells and 620,000 listing outcomes. These outcome rows are repeated predictions, not independent evaluation examples.

Table 7: Primary descriptive comparison: robust frozen estimator, 10% capacity. Mean worst unilateral log-quote gain, equally averaging five fixed peer profiles and three seeds. No confidence interval or practical-threshold pass is asserted.

Source	Rule	Original	Capped	Reduction (%)
jucars	benefit	0.010473	0.009736	7.04
jucars	cdm	0.010389	0.010041	3.36
jucars	risk	0.010748	0.010748	0.00
jucars	suppression	0.010546	0.010007	5.11
mucars	benefit	0.007412	0.006486	12.49
mucars	cdm	0.006780	0.006399	5.62
mucars	risk	0.006540	0.006540	0.00
mucars	suppression	0.007607	0.006984	8.18
synthetic	benefit	0.011395	0.010699	6.11
synthetic	cdm	0.011560	0.011244	2.74
synthetic	risk	0.011423	0.011423	0.00
synthetic	suppression	0.012591	0.012085	4.01

The design contains 1,000 unique evaluation listings in ten shared batches of 100, with ten fitting-seed realizations on fixed training records. JUCars and MUCars are reported separately below; the completed union covers all thirty source/seed pairs.

At 10% capacity, SAV mean positive log-quote inflation is 0.021175 versus 0.012084 for static robust without feedback, a 75.23% increase. Table 8 retains all five original contrasts. The benefit-head variant is the original feedback-trained variant; it is not the separately tuned ordinary-benefit benchmark. This extension does not isolate estimator, acquisition and feedback factors, and it must not be presented as the matched factorial experiment.

Original comparator	Comparator minus SAV	Nominal 95% interval
Benefit head	−0.004328	$[-0.007774, -0.001215]$
Static robust	−0.009091	$[-0.012355, -0.006323]$
Masking benefit	0.029153	$[0.024928, 0.033582]$
SAV without gap	0.017007	$[0.014282, 0.019794]$
Pre-audit attacker	−0.007556	$[-0.010444, -0.004923]$

Table 8: Synthetic, original protocol, ten fitting seeds and 10% capacity. Positive paired inflation differences favor SAV. Intervals are exploratory, unadjusted percentile intervals from 1,000 crossed seed/batch resamples.

For each contrast we construct the ten-seed by ten-batch paired difference matrix. We independently resample seed indices and shared batch indices, retaining their crossed grid and within-cell method pairing. These intervals capture empirical fitting-seed and evaluation-batch variation conditional on the training records; they do not capture training-population uncertainty. Only ten batch clusters are available, so nominal coverage is not established. No multiplicity correction or confirmatory hypothesis-test claim is made. An independent checker reconstructs all five contrasts and bootstrap endpoints from raw outcome records using count-weighted resampling, and rejects altered effect direction, interval endpoints and missing-seed inputs.

C.2 JUCars and sensitivity to seed variability

Takeaway. The original-protocol JUCars result does not support a stable SAV advantage, and its uncertainty depends materially on including fitting-seed variation.

Table 9 retains the five primary comparisons. The same ten-seed source-binding and recorded-outcome audit covers 620 JUCars method/capacity cells and 620,000 outcome rows, again representing 1,000 unique evaluation listings in ten shared batches. All five raw-record contrasts and bootstrap endpoints pass independent reconstruction. At 10% capacity, SAV inflation is 0.014988 versus 0.012082 for static robust, a 24.05% increase. The comparator-minus-SAV difference is -0.002906. Its fixed-seed batch-only interval is $[-0.003783, -0.002103]$, whereas the crossed seed/batch interval is $[-0.007025, 0.000777]$. The latter includes zero. This is uncertainty about a paired effect, not evidence of equivalence or a demonstrated source interaction.

Original comparator	Comparator minus SAV	Nominal 95% interval
Benefit head	0.001252	$[-0.000172, 0.002946]$
Static robust	-0.002906	$[-0.007025, 0.000777]$
Masking benefit	0.005030	$[0.001649, 0.007985]$
SAV without gap	0.019588	$[0.016131, 0.023292]$
Pre-audit attacker	-0.001037	$[-0.002428, 0.000303]$

Table 9: JUCars, original protocol, ten fitting seeds, 10% capacity. Positive differences favor SAV. Crossed seed/batch percentile intervals use 1,000 resamples, are exploratory and unadjusted, and condition on fixed training records. The limitations stated for the synthetic extension apply.

C.3 MUCars completes the source comparison

Takeaway. The original-protocol MUCars extension does not establish a stable advantage over static robust or the original benefit head. This comparison differs from the separately tuned ordinary-benefit benchmark.

Table 10 retains the five primary comparisons. The completed three-source union contains thirty source/seed pairs, 1,860 method/capacity cells and 1,860,000 repeated outcome rows. Each source still contains only 1,000 unique evaluation items. An independent raw-record checker reconstructs all fifteen paired contrasts and both bootstrap definitions. At 10% capacity, MUCars SAV inflation is 0.007340 against 0.006511 for static robust, a 12.73% increase. The crossed interval includes zero; this neither establishes superiority nor demonstrates equivalence. Against the original benefit head, SAV inflation is 3.53% higher, also with an interval spanning zero. The large gains against masking benefit and SAV without gap remain, illustrating why comparator identity must accompany any statement that MUCars favors SAV.

D Conditional probability-stability diagnostics

Takeaway. Randomized allocation admits a conditional sensitivity bound, but small field margins and scores near zero limit its applicability. Stability alone does not imply lower strategic inflation.

Fix the positive-score support I, the queried field of each item, and budget B. For two score vectors g, g' with all supported scores at least $m > 0$, let $\delta = \|g - g'\|_\infty$. Define the score-minimax value and canonical marginals

$$T(g) = \min_{0 \le p \le 1,\, \sum_i p_i \le B} \max_{i \in I} g_i(1 - p_i), \qquad p_i(g) = [1 - T(g)/g_i]_+.$$

Original comparator	Comparator minus SAV	Nominal 95% interval
Benefit head	−0.000250	[−0.001092, 0.000488]
Static robust	−0.000829	[−0.001870, 0.000210]
Masking benefit	0.039742	[0.036708, 0.042906]
SAV without gap	0.015624	[0.014036, 0.017120]
Pre-audit attacker	0.000580	[−0.000212, 0.001238]

Table 10: MUCars, original protocol, ten fitting seeds, 10% capacity. Positive differences favor SAV. Crossed seed/batch percentile intervals use 1,000 resamples and retain the fixed-training-record, exploratory and unadjusted interpretation of the other two source extensions.

For $B = 0$ use zero marginals; for $B \geq |I|$ use unit marginals on I. Then

$$|T(g) - T(g')| \leq \delta, \qquad \|p(g) - p(g')\|_\infty \leq \min\{1, 2\delta/m\}.$$

For any fixed feasible allocation, the two objectives differ by at most δ, which proves the value bound after minimization. The function $h(t, x) = [1 - t/x]_+$ is separately $1/m$-Lipschitz in t and x for $t \geq 0, x \geq m$: on its positive branch, $t/x^2 \leq 1/x$, and it is continuous at the branch boundary. Applying the value bound proves the marginal bound. The active allocation set may change; positive-score support and queried fields must remain fixed.

If both pre-query and recovered quotes change by at most ϵ, and the primed recovery span is at most R, expected quotes differ by at most $\epsilon + R \min\{1, 2\delta/m\}$. If these conditions hold uniformly over all own reports at fixed peer reports, costs are unchanged, and truthful quotes are verification-neutral, the positive unilateral-gain envelope changes by at most $2\epsilon + R \min\{1, 2\delta/m\}$. This is not a guarantee for adapted peer responses, equilibrium selection, training convergence, or a fixed realization of the sampling randomness.

Three examples delimit the claim. With two items and $B = 1$, swapping scores $(1 + \delta, 1)$ and $(1, 1+\delta)$ changes the first item's top-B probability by one, but its minimax probability by $\delta/(2+\delta)$. In contrast, swapping $(2\delta, \delta)$ and $(\delta, 2\delta)$ changes its minimax probability by $1/3$ even as δ tends to zero. Finally, one fully queried item with two fields whose scores swap as in the first example can have a unit quote jump if the two recovery quotes are zero and one. The score lower bound and fixed-field condition therefore cannot simply be removed. An exact-arithmetic checker covers 2,916 score-pair/budget cases, including 510 active-set changes; 108 production/linear-program comparisons agree to floating-point precision. These finite checks supplement the proof.

We additionally audited all 1,080 saved fixed-profile batch executions at hypothetical radii $\delta = \alpha \,\mathrm{median}(g_i : g_i > 0)$. A sufficient certificate requires support preservation and every positive item's top-two field gap to exceed 2δ. Its bound uses $m = \min_{i \in I} g_i - \delta$. Recovery spans are reconstructed for every eligible item, including items not queried by a saved top-B allocation. Table 11 records the sufficient-certificate counts. Each row has 1,080 batch executions, including both generating allocators; these are dependent diagnostic cases, not independent statistical replicates. Top-B is locally constant when the same support/field conditions and its selection-rank margin hold (or all positive items fit). Its certificate is stronger than a nonzero sensitivity bound, so counts do not establish an ordering of stability. Failure to certify is not evidence of instability. An independent checker reconstructs all 4,320 condition rows and 144 grouped summaries using exact fractions of the stored scores and radii; displayed quote bounds remain floating-point approximations. The radii are not calibrated prediction errors. These post-hoc diagnostics neither establish performance improvement nor explain source effects causally. General water-filling sensitivity is established literature; priority for this application-specific bound is not claimed.

α	Minimax conditions	Probability bound < 1	Top-B constant
0.001	471	468	401
0.01	401	353	212
0.05	267	245	53
0.1	244	242	39

Table 11: Post-hoc stability certificates at four hypothetical relative score radii. Each row covers the same 1,080 dependent fixed-profile batch executions. Counts certify sufficient conditions, rather than observed perturbation errors or a ranking of performance.

E Strong additional comparators and adaptation boundaries

The following controls belong to earlier independently audited experiments, not the newly trained matched factorial model family. Keeping these identities separate prevents a favorable acquisition or tuning result from being attributed to the wrong training intervention.

E.1 Ordinary-benefit and equal-allowance controls

A robust/no-feedback price model with ordinary error-reduction acquisition was evaluated on all six budgets and the original three seeds. The independently checked 54-cell extension retained identical honest quotes, public scoring, allocations and first-batch exact deviations. At 10% capacity, reference inflation was 0.01077, 0.01117 and 0.00778 on synthetic, JUCars and MUCars, respectively. Reference-minus-SAV differences were -0.00998, -0.00432 and 0.00089, with conditional shared-batch intervals $[-0.01260, -0.00791]$, $[-0.00598, -0.00285]$ and $[0.00045, 0.00130]$. These intervals averaged the three fitted seeds before resampling batches; they are not the ten-seed crossed intervals in the main factorial analysis.

An additional size/tuning study retained training caps 1,000, 2,500 and 5,000, three fitting seeds, three training-internal gap-weight trials and twelve residual epochs for both candidates. Its 972 parent tuning cells and 324 ordinary-benefit reference cells passed full audits. At the largest cap, tuned SAV inflation was 20.41% lower on MUCars but 30.09% higher on synthetic and 27.09% higher on JUCars than the separately tuned ordinary-benefit reference. The associated reference-minus-SAV intervals were $[0.00097, 0.00186]$, $[-0.00682, -0.00204]$ and $[-0.00333, -0.00125]$. All scheduled caps and fixed/tuned roles remain in the archived evidence; no favorable training size replaces the largest-cap comparison. This further supports conditional gains rather than a consistent candidate advantage.

E.2 Categorical support-margin imputation

A whole-field adaptation of support-margin imputation (SMI) [9] used all allowed covariates and four optional-field groups. One deterministic fitting generation per source trained a sixteen-model linear support-vector regression family on complete-optional-field training rows. Scaling, category vocabularies and target normalization were training-only. Missing categories required a one-hot completion including an unknown token. If the reduced sign-consistent model family or its optimization was unavailable, the method used ordinary median/mode imputation. The original SMI guarantee is not asserted on these fallback rows, and this is not an unchanged official reproduction.

An independent categorical-enumeration and scalar-interval auditor checked 94,527 report/domain cases and every final quotation and legal no-query report. Training cohort sizes were 2,947, 2,674 and 3,368 for synthetic, JUCars and MUCars. Across unrestricted, nonnegative and

unit-box completion domains, aggregate outcomes matched ordinary median/mode regression. Attacked fallback fractions were 75.3%, 87.3% and 65.6%, respectively. This diagnoses a limit of the categorical adaptation's retained-family assumptions; it does not refute the original SMI theorem. These no-query fits differ in cohort and model class from the matched factorial predictor, so their higher inflation cannot isolate a benefit of the present acquisition procedure.

E.3 Acquisition-conditioned neighbor control

ACO-1 adapted acquisition-conditioned expected-loss scoring [12] to one-field regression queries and batch capacity. A training-only neighbor bank estimated loss changes for sixteen observation patterns; unavailable recorded bank values gave zero replay benefit. Distances used observed query groups and permitted nonoptional features, with fixed category/numeric weights and stable ties. It was a clean-room task adaptation, not an unchanged execution of the original sequential classification method.

The independently audited nine-pair study contained 144 cells. The declared main $k = 25$ used every capacity; fixed $k = 10$ and $k = 50$ sensitivities used 10% capacity only. No winning k was selected. With the frozen robust predictor and $k = 25$, mean inflation was 0.01081 on synthetic, 0.01118 on JUCars and 0.00716 on MUCars. Full-SAV-minus-ACO differences of 0.00995 and 0.00431 on synthetic and JUCars had conditional intervals $[0.00725, 0.01345]$ and $[0.00291, 0.00588]$. On MUCars the point comparison favored SAV by 0.00027, but the interval crossed zero. Thus a strong predictive acquisition control can outperform the original candidate. These results strengthen comparator coverage without implying that one-step ACO-1 dominates the matched factorial policies or reproduces the original method's guarantee.

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